Finding Zeros with the Rational Root Theorem

Testing candidates, factoring completely, and finding every zero. For a polynomial with integer coefficients, every rational zero has the form $\pm\frac{p}{q}$, where p divides the constant term and q divides the leading coefficient. The routine is always the same: **(1)** list the candidates, **(2)** test them until one gives a remainder of 0, **(3)** divide it out, **(4)** finish the depressed quotient by factoring or by the quadratic formula. Problems 1–2 walk that routine on a single polynomial; after that you run it yourself. Show your candidate list and your division work — both are part of the answer.

1. Let $f(x) = x^3 - 4x^2 + x + 6$. The rational root candidates are ± 1, ± 2, ± 3, ± 6. Evaluate $f(2)$ to decide whether $x = 2$ is a zero of f.

Answer: __________________

2. Using your result from problem 1, divide out the corresponding factor and find *all* zeros of $f(x) = x^3 - 4x^2 + x + 6$.

Answer: __________________

3. Factor $g(x) = x^3 + 2x^2 - 5x - 6$ completely over the integers.

Answer: ____________________

4. Let $h(x) = 2x^3 - x^2 - 8x + 4$. Because the leading coefficient is 2, the candidate list includes halves. Evaluate $h\left(\frac{1}{2}\right)$ to decide whether $x = \frac{1}{2}$ is a zero.

Answer: ____________________

5. Find all zeros of $h(x) = 2x^3 - x^2 - 8x + 4$.

Answer: ____________________

6. Find all zeros of $p(x) = x^3 - 4x^2 + 2x + 4$. Exactly one zero is rational — find it first, then solve the quadratic that is left. Give exact answers, not decimals.

Answer: ________________

7. Factor $q(x) = 3x^3 - x^2 - 12x + 4$ completely over the integers.

Answer: ________________

8. Solve $x^3 - x^2 - 13x - 3 = 0$. Give all real solutions in exact form.

Answer: ________________

9. Find all zeros of $r(x) = x^4 - 2x^3 - 7x^2 + 8x + 12$. A quartic needs the routine twice: after the first division you are left with a cubic, so run the theorem again on it.

Answer: ________________

10. Find all zeros of $s(x) = x^3 - x^2 + 4x - 4$, including any non-real zeros. State how many zeros the theorem itself was able to supply.

Answer: __________________

11. Factor $t(x) = 4x^3 - 8x^2 - x + 2$ completely over the integers.

Answer: __________________

12. **Synthesis.** For $u(x) = x^3 - 3x - 1$ the candidate list is just $x = 1$ and $x = -1$, and $u(1) = -3$ and $u(-1) = 1$, so neither is a zero. A classmate concludes: "The rational root theorem found nothing, so u has no real zeros." Explain in two or three sentences why that conclusion is wrong, and give evidence that u does have a real zero between $x = -1$ and $x = 1$.